Nature of phase transitions and metastability in scalar-tensor theories

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Compact stars above a critical stellar mass develop large scalar fields in some scalar-tensor theories. This scenario, called spontaneous scalarization, has been an intense topic of study since it passes weak-field gravity tests naturally while providing clear observables in the strong-field regime. The underlying mechanism for the onset of scalarization is often depicted as a second-order phase transition. Here, we show that a first-order phase transition is in fact the most common mechanism. This means metastability and transitions between locally stable compact object configurations are much more likely than previously believed, opening vast new avenues for observational prospects.

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Advances in gravitational and electromagnetic observations now enable us to test general relativity (GR) in unprecedented ways [1–7]. The particular scenario of *spontaneous scalarization* in scalar-tensor theories (STTs) has been a prime target of study for deviations from GR, as it readily satisfies the strict observational limits in weak fields while also promising achievable observational targets in strongly gravitating systems [8, 9]. Hence, the resulting literature has been vast and impactful [10–18]. Here, we report that a first-order phase transition is a considerably more likely mechanism for the onset of scalarization, unlike the commonly invoked second-order phase transition picture. This means distinct phenomena such as metastable neutron star (NS) configurations and discontinuous jumps between them are more likely than previously thought. This, in turn, opens new avenues for both theory and observation.

In spontaneous scalarization, gravity is governed by a fundamental scalar field in addition to the metric tensor, and the scalar field vacuum can become unstable in the highly curved regions of spacetime [9]. While ordinary stars and low-mass NSs can behave as GR dictates, a scalar cloud forms around the NS if its mass exceeds a critical value. This is commonly explained in analogy to spontaneous magnetization, and we have a *second-order phase transition* at the critical mass [10]. The scalar field strength continuously rises from zero with increasing stellar mass, and can have large enough values to cause nonpertubative deviations from GR that are relatively easy to observe.

We show that the above picture only holds for a very limited part of the STT parameter space, and the onset of scalarization occurs as a discontinuous jump in the stellar structure in most cases. This means scalarized and unscalarized solutions are both possible for some stellar masses, one of the solutions being the globally stable one while the other is metastable. All these findings are successfully explained as a first-order phase transition, which also elucidates many previously opaque numerical results.

Metastability in scalarization has been investigated in some particular contexts before [18–23], but our results imply that first-order phase transitions are the rule rather than the exception. This means limited existing studies have much wider applicability than previously considered, providing novel ways to test gravity.

The theory: Consider the STT action

$$S = \frac{1}{16\pi}\int \mathrm{d}^4x\sqrt{-g}\left(R - 2g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - 2m_\phi^2\phi^2\right) + S_\mathrm{m}[f_\mathrm{m}, \tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}]. \quad (1)$$

S_m is the matter action and f_m denotes the matter fields which couple to the conformally scaled *Jordan-frame metric*. We use $A(\phi) = e^{\beta\phi^2/2}$ as in the original discovery paper of Damour and Esposito-Farèse [8], and add the scalar mass term m_ϕ [24]. Hence, our STTs live on a (β, m_ϕ) parameter space. Note that m_ϕ has the important consequence of suppressing the scalar dipole radiation [9, 24, 25], which means our model here is not affected by various observations that severely constrain the original massless scalarization idea [26–28].

The scalar field obeys

$$\Box\phi = \left(-4\pi\beta A^4\tilde{T} + m_\phi^2\right)\phi, \quad (2)$$

$\tilde{T}$ being the trace of the stress-energy tensor in the Jordan frame. The term in the parentheses becomes negative at high matter densities for $\beta < 0$, which leads to a mode with purely imaginary oscillation frequency, a tachyon. Scalar fields grow exponentially around massive NSs due to this instability, which is eventually suppressed by nonlinear effects, leading to a star inside a stable scalar cloud, called a *scalarized star*. Note that an unscalarized star with $\phi = 0$, a GR solution, is always an equilibrium solution for our choice of $A(\phi)$, but it is not stable if the tachyon is present.

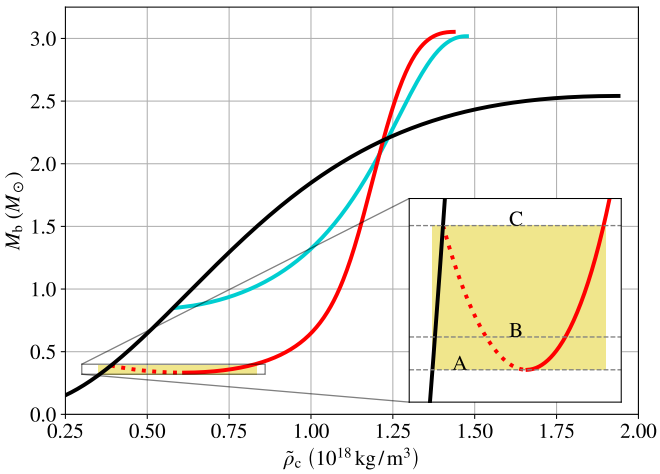

FIG. 1. Baryon mass vs the central density $\tilde{\rho}_c = \tilde{\rho}(r{=}0)$ diagrams under general relativity (black) and STTs with spontaneous scalarization (red, turquoise). Turquoise: $\beta = -10$, $m_\phi = 1 \times 10^{-11}$ eV. Red: $\beta = -40$, $m_\phi = 4 \times 10^{-11}$ eV.

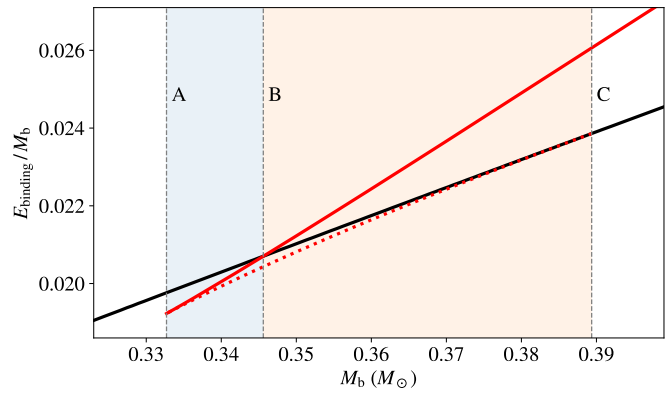

FIG. 2. Fractional binding energy $(M_b - M_{\rm ADM})/M_b$ in the highlighted region of Fig. 1.

Metastability in spontaneous scalarization: We used the numerical techniques detailed in Tuna *et al.* [29] to obtain the static, spherically symmetric NS solutions for the action (1), also see the "End Matter." Both scalarized and unscalarized solutions are plotted in Fig. 1 as one-parameter families in terms of the central density of the star $\tilde{\rho}_c$. We use the HB equation of state (EOS) of Read *et al.* [30] in all cases, but results are qualitatively similar for other choices.

Our main objective is understanding the physical relevance of the solutions when there exists more than one configuration for a given number of baryons N_b in the star, or equivalently the total baryon mass $M_b = m_b N_b$, where m_b is the rest mass of a single baryon. In other words, if we are given a fixed number of baryons and there are more than one theoretical equilibrium configuration they can arrange themselves in, which one(s) are astrophysically relevant?

Let us start with the "standard" scalarization picture. The unscalarized solutions, which are also solutions in GR, are on the black curve, and the scalarized ones are on the turquoise one in Fig. 1. Scalarized solutions branch off at a critical baryon mass $M_{\rm crit}$, below which there is only one solution for a given baryon mass; an unscalarized one. This solution is known to be stable, hence, can be observed in principle. For $M_b > M_{\rm crit}$ though, there are two possible solutions, one of them being scalarized [31]. However, we have seen that the unscalarized solution is unstable to the growth of a scalar mode, hence it would not be observed. Thus, any existing star with these higher masses would be a scalarized one. We call this picture *second-order scalarization.* The naming will become clear in our phase transition discussion.

Our main interest is the *first-order scalarization* of the red curve in Fig. 1, which corresponds to different theory parameters (β, m_ϕ). The scalarized section of the $M_b(\tilde{\rho}_c)$ curve slopes downwards at the branch-off point $M_{\rm crit}$ in this case, which changes the above discussion substantially. Defining the lowest mass on the scalarized branch as $M_{\rm bottom}$, there is a single scalarized and stable equilibrium solution for $M_b < M_{\rm bottom}$. There are two solutions for $M_b > M_{\rm crit}$, where the unscalarized solution (black) is unstable due to the tachyonic growth, and the scalarized one (red) is stable. This is similar to second-order scalarization so far.

The novelty is in the highlighted region $M_{\rm bottom} < M_b < M_{\rm crit}$ in Fig. 1, where there are three equilibrium solutions for a given M_b. Firstly, the middle solution on the dotted curve, is unstable, hence physically irrelevant. A one parameter family of equilibrium NS solutions, such as our $M_b(\tilde{\rho}_c)$, change from being stable to unstable at a turning point $\mathrm{d}M_b/\mathrm{d}\tilde{\rho}_c = 0$, and the $\mathrm{d}M_b/\mathrm{d}\tilde{\rho}_c < 0$ part is known to be the unstable one [14, 32–35]. This instability is not necessarily due to the tachyonic scalar mode, but can also be a result of the modes of collective motion of the stellar matter.

The remaining two solutions, the scalarized star with $\mathrm{d}M_b/\mathrm{d}\tilde{\rho}_c > 0$ (solid red) and the unscalarized star (black), are at least locally stable. The former is stable by being on the other side of a turning point, and the latter because there is no hydrodynamical instability on this part of the GR curve, and the tachyonic instability of the scalar is not present before $M_{\rm crit}$. Thus, we have two physically relevant solutions sharing the same M_b.

When there are multiple equilibrium solutions that are robust against small perturbations, as is our case, we have the phenomenon of *metastability.* We typically call the solution with the globally lowest energy *the* stable one (or the global minimum), whereas the others are metastable. A metastable state would energetically prefer to transition to the global minimum, but this can only occur for large enough perturbations which may or may not exist in a star's environment. Hence, both solutions are possible in reality, which makes novel astrophysical scenarios possible as we will discuss later.

The total energy of a NS is its ADM mass $M_{\rm ADM}$. Hence, we can determine metastability by checking the total binding energy $E_{\rm binding} = M_{\rm b} - M_{\rm ADM}$ plotted in Fig. 2, which shows that metastability is not exclusive to one branch of solutions. For lower $M_{\rm b}$ (to the left of the vertical line B) the unscalarized (black) configuration is energetically favored, and the opposite is true for higher $M_{\rm b}$. We also see that the unstable scalarized solutions (dotted red) always have higher energy than the locally stable ones.

We closely investigated only two points on the (β, m_ϕ) parameter space of our STT so far, but what is the prevalence of first- and second-order scalarization in general? Fig. 3 (left) shows the maximum ADM mass difference ΔM between two stars with the same baryon mass, which is a crude approximation for the energy that would be released in a transition between two states where there is no matter loss. Since it means there are metastable states, any point with a nonzero value means scalarization is first order if the STT with parameters (β, m_ϕ) governs gravity. One can see that first-order scalarization is the dominant type even for the original scalarization model for which $m_\phi = 0$ [8]. This clearly demonstrates our claim about the ubiquity of first-order scalarization.

Fig. 3 (right) shows the ADM mass at the branch-off point of scalarization, $M_{\rm ADM,branch}$ only for the parameter values featuring first-order scalarization. This is relevant since it gives us a rough measure of the NS (ADM) masses where transitions from a metastable to a stable configuration can occur. Metastable solutions mostly have $M_{\rm ADM,branch} \lesssim 0.8 M_\odot$, which is low but astrophysically relevant [36]. We will further comment on this in our discussion.

In terms of detectability, relatively higher NS masses and higher ΔM are both desirable. This means that the prominent regions in Fig. 3 are not necessarily interesting on their own, but rather the middle region where ΔM and $M_{\rm ADM,branch}$ are both relatively high likely provides the best observational prospects. Consider these results in the context that they are only for a specific STT. ΔM and $M_{\rm ADM,branch}$ would vary for other, more general, STTs, providing opportunities for testing deviations from GR.

Phenomenological descriptions of scalarization: The onset of scalarization has been viewed as a phase transition since the early days [10], and phenomenological explanations in terms of Landau theory have been occasionally invoked in the literature [35, 37]. Nevertheless, this has been in the context of second-order phase transitions, which only holds for second-order scalarization [38]. Here, we show that all unusual features of first-order scalarization in the $M_{\rm bottom} < M_{\rm b} < M_{\rm crit}$ region can be explained by a first-order phase transition.

In Landau theory [39, 40], one considers how the total free energy of a system changes with a quantity called the *order parameter*, based on the symmetries of the system.

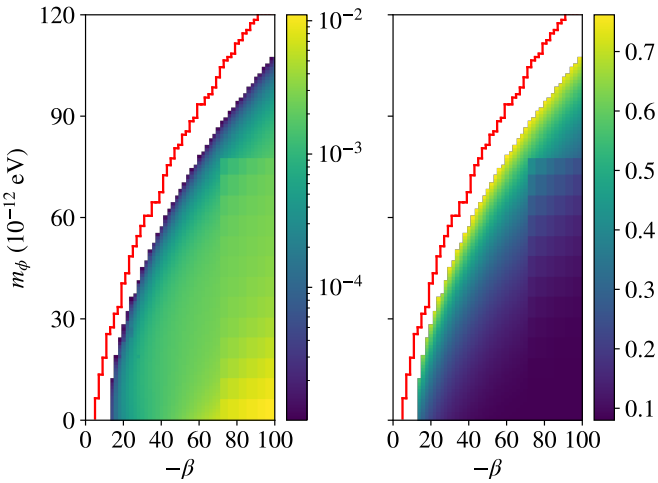

FIG. 3. Dependence of scalarization characteristics on the theory parameters (β, m_ϕ) for first order scalarization. There is no scalarization at all above the red line [29]. Second-order scalarization occurs only in the white regions below the red line. Thus, first-order scalarization is by far the more likely outcome if scalarization occurs. Left: ΔM, the maximum ADM mass difference between two stars of equal baryon mass. Right: ADM mass at the scalarization branch-off point. Both are in units of $M_\odot$. $\Delta M < 10^{-5} M_\odot$ is indistinguishable from numerical noise, hence excluded from the plot. We use lower resolution away from the boundary between first- and second-order scalarization, where fine detail is not critical. See "End Matter" for details.

In our case, we will concentrate on the simpler $m_\phi = 0$ case where this translates to

$$M_{\rm ADM} = M_0(M_{\rm b}) + a(M_{\rm b})Q^2 + \frac{1}{2}b(M_{\rm b})Q^4 + \frac{1}{3}c(M_{\rm b})Q^6. \tag{3}$$

The *scalar charge* of the star, defined as $\phi(r \to \infty) = \phi_\infty + Q/r + \cdots$ has been used as the order parameter Q [8, 10][41]. In rough terms, we take a given amount of baryons $M_{\rm b}$, and determine how the total energy of the solution changes as we dress it with scalar fields whose strength is represented by Q. The scalar charge is not well defined for massive scalars, but some other measure of the strength of scalarization such as the scalar field value at the center can be used in a similar vein. Only even terms are present in Eq. (3) due to the $\phi \to -\phi$ symmetry. $c(M_{\rm b}) > 0$ for overall stability of the system.

Second-order scalarization occurs when $b(M_{\rm b}) > 0$. We overlook $c(M_{\rm b})$ which plays no direct role in this case. The two variables $M_{\rm b}$ and Q are actually not independent for the equilibrium solutions, i.e., the static NS solutions we studied above. For a given $M_{\rm b}$, the physically realized Q will be the one that minimizes the total energy $M_{\rm ADM}$. For $a > 0$, this is simply $Q = 0$, the unscalarized solution. For $a < 0$, however, $Q = 0$ becomes a local maximum, an unstable equilibrium, and we get two minima at $Q = \pm(-a/b)^{1/2}$. Thus, as a changes sign, the $Q \leftrightarrow -Q$ symmetry gets spontaneously broken, and a phase transition to nonzero values of Q occurs *con-*